PART-1 : PHYSICS

SECTION–I : (Maximum Marks : 32)

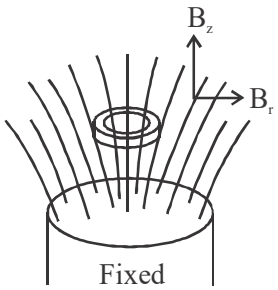
- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –1	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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1. Figure shows non-uniform magnetic field developed by magnetic rod. The magnetic field is described in terms of $B_r = 0.3r, B_2 = (1-2z)$ where r, z are radial and vertical position coordinates respectively. A resistance less ring of mass m radius a, inductance L is shown in figure. Initially centre of ring is present at r = 0, z = 0 and Current flowing in it is zero. (g is acceleration due to gravity)



(A) Current through ring is $\frac{2z\pi a^2}{L}$

(B) Motion of ring is periodic but not S.H.M

(C) Maximum z coordinate of centre of ring during its motion is $\frac{mgL}{0.6\pi^2 a^4}$

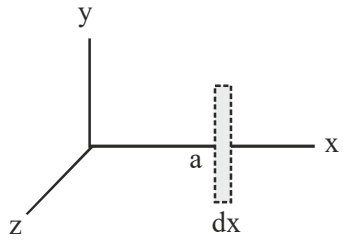
(D) Amplitude of the motion is $\frac{mgL}{1.2\pi^2 a^4}$

Ans. AD

Sol. $\Delta\phi = 0$

$F_{net} = mg - 0.3a2\pi ai$

2. A plane electromagnetic sinusoidal wave propagating in the +x direction. At x = a electric field is $E_0(-j)$ and at x = a + dx it is $(E_0 + dE)(-j)$ then (B – magnetic field)



(A) Direction of B in shaded region is $(-\hat{k})$ and rate of change of B with respective time is +ve

(B) Direction of B in shaded region is $(+\hat{k})$ and rate of change of B with respective time is –ve

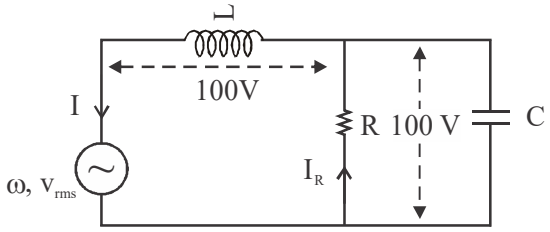
(C) Direction of B in shaded region is $(+\hat{k})$ and rate of change of B with respective time is +ve

(D) Direction of B in shaded region is $(-\hat{k})$ and rate of change of B with respective time is –ve

Ans. D

Sol. $\frac{\partial E}{\partial x} = -\frac{\partial B}{\partial t}$

3. In shown circuit rms voltage across inductor and resistor is 100 V



(A) V_{rms} can be any value between 0 to $100\sqrt{2}V$

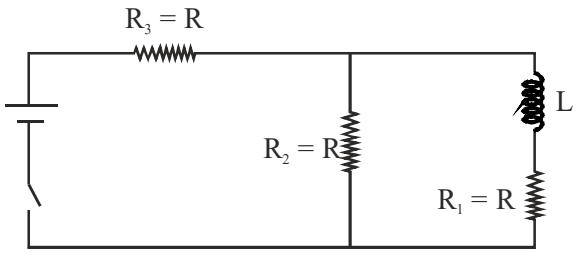
(B) If V_{rms} = 100 V and $I_R = \frac{5\sqrt{3}}{2}A$ then I = 5A

(C) If $V_{rms} = 100V$ and $I_R = \frac{5\sqrt{3}}{2}A$ then capacitive reactance is 40Ω

(D) V_{rms} can be any value between 0 to 200 V

Ans. ABC

4. A circuit shown in figure contains ideal inductor and battery. Initially current through inductor is zero and the switch is closed at t = 0. When potential difference across inductor is 50% of maximum potential difference across it then select correct alternative/s:



(A) Current through battery is 7 E/12 R
(B) Current through resistance R_2 is 5E/12R
(C) Current through inductor is E/6R
(D) Potential difference across R_1 is E/2

Ans. ABC

Sol. $i_L = \frac{E}{3R}\left(1 - e^{-t3R/2L}\right)$

$V = \frac{EL}{3R} e^{-\frac{t32}{2L}} \left(\frac{3R}{2L}\right) = \frac{E}{2} e^{-\frac{3RT}{2L}}$

At $t = t_0$ $V = E/4$

$e^{-3Rt_0/2E} = \frac{1}{2}$

$i_L = \frac{E}{3R}\frac{1}{2} = \frac{E}{6}R$

$V_{R_1} = \frac{E}{6}$ $V_{R_2} = \frac{E}{6} + \frac{E}{4} = \frac{5E}{12}$

5. The two slits of double slit experiment were moved symmetrically apart with small relative velocity v and the distance between screen and mid-point of slits is fixed and equal to D. Consider a point P on the screen at a distance x from central maxima then (x < < < D):

(A) Rate of change of number of fringes between central maxima and point P changes with respect to time is $\frac{xv}{\lambda D}$.
(B) Number of fringes contained between central maxima and point P increases with time.
(C) Fringe width decreases as time passes.
(D) Maxima's are moving closer to each other.

Ans. ABCD

6. In the nuclei X decay series begins with 232X. (Here 232 is mass number).
228X alpha - decays into 224 Y with a half of 10000 seconds, 224Y alpha-decays into 220Z with a half life of 100 seconds, which in turn decays into 216W with a half life of 50 s. If the nuclides are in radioactive equilibrium in a mineral sample that contains 1.0 g of 220Z. At this instant

(A) Mass of 232 Th nearly is 310 g
(B) Mass of 220 Rn nearly is 2g
(C) Mass of 232 Th nearly is 200 g
(D) Mass of 220 Rn nearly is 200 g

Ans. BC

Sol. $\frac{m_1}{A_1T_1} = \frac{m_2}{A_2T_2} = \frac{m_3}{A_3T_3}$

$\frac{m_2}{224 \times 100} = \frac{10^{-3}}{220 \times 50}$

$m_2 = \frac{224}{110} \times 10^{-3} = 0.002$

7. The activity of a sample of radioactive material is R_1 at time t_1 and R_2 at time $t_2 (t_2 > t_1)$. If mean life of radioactive sample is T, then

(A) $R_1 t_1 = R_2 t_2$ (B) $\frac{R_1 - R_2}{t_2 - t_1} = \text{constant}$

(C) $R_2 = R_1 \exp\left(\frac{t_2 - t_1}{T}\right)$ (D) $R_2 = R_1 \exp\left(\frac{t_1}{T t_2}\right)$

Ans. C

8. A moving H-atom makes a head on collision with a stationary Li++ ion. Just before collision, both H and Li ++ are in their first excited state. Immediately after collision H was found to be in ground state and Li ++ was in second excited state. No photon is emitted in the process. Assume that mass of Li++ is nearly 7 times the mass of H.

(A) Initial kinetic energy of H-atom can be 7 eV
(B) Initial kinetic energy of H-atom can be 10 eV
(C) Initial kinetic energy of H-atom can be 6 eV
(D) Initial kinetic energy of H-atom can be 8.7 eV

Ans. BD

SECTION–IV : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II.**
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II.**
- One or more entries in **Column-I** may match with one or more entries in **Column-II.**
- The ORS contains a 4 × 5 matrix whose layout will be similar to the one shown below :

(A) (P) (Q) (R) (S) (T)
(B) (P) (Q) (R) (S) (T)
(C) (P) (Q) (R) (S) (T)
(D) (P) (Q) (R) (S) (T)

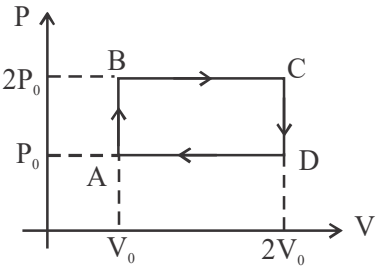
- For each entry in **column-I,** darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks	:	+2	If only the bubble(s) corresponding to all the correct match(es) is (are) darkened
Zero Marks	:	0	In none of the bubbles is darkened
Negative Marks	:	0	In all other cases

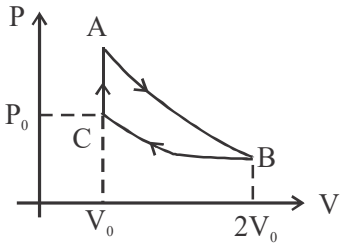
9. **Column I** **Column II**

(A) An ideal mono atomic gas undergoes the following cyclic process.



(P) Work done in process A to B is 0

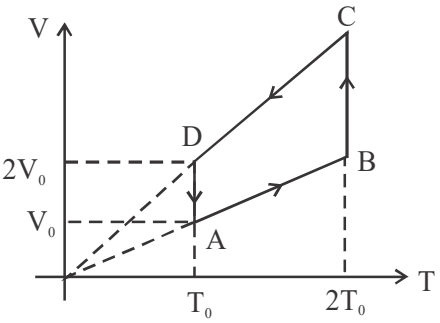
(B) An ideal polyatomic non-linear gas undergoes the cyclic process shown in (Take : $2^{4/3} = 2.5$)



$A \rightarrow B$ (Adiabatic)
$B \rightarrow C$ (Isothermal)
$C \rightarrow A$ (Isochoric)

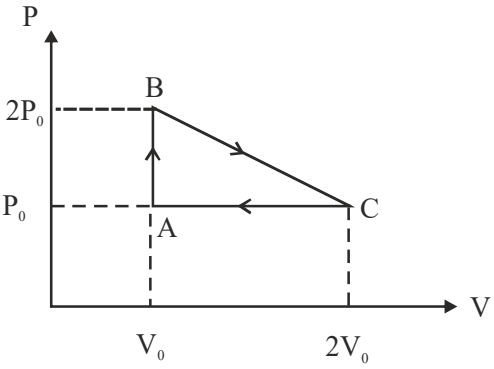
(Q) Change in internal energy from B to C is 0

(C) An ideal monoatomic gas undergoes the following process.



(R) Efficiency of the cycle is less than 0.5.

(D) An ideal mono atomic gas undergoes the following cyclic process. The pressure at point A is P_0

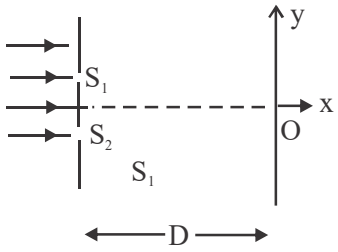


(S) Work done in process A to B is $(3/4)P_0V_0$

(T) Heat supplied to the gas in process A to B is positive

Ans. A $\rightarrow$ P, R, T
B $\rightarrow$ Q, R, S, T
C $\rightarrow$ Q, R, T
D $\rightarrow$ P, Q, R, T

10. A monochromatic parallel beam of light of wavelength λ is incident normally on the plane containing slits S_1 and S_2. The slits are of unequal width such that intensity only due to one slit on screen is four times that only due to the other slit. The screen is placed perpendicular to x-axis as shown. The distance between slits is d and that between screen and slit is D. Match the statements in Column - I with results in Column - II.



Column - I	**Column - II**
(A) The distance between two points on screen having equal intensities, such that intensity at those points is $\frac{1}{9}$th of maximum intensity	(P) $\frac{D\lambda}{3d}$
(B) The distance between two points on screen having equal intensities, such that intensity at those points is $\frac{3}{9}$th of maximum intensity	(Q) $\frac{D\lambda}{d}$
(C) The distance between two points on screen having equal intensities, such that intensity at those points is $\frac{5}{9}$th of maximum intensity	(R) $\frac{2D\lambda}{d}$
(D) The distance betwen two points on screen having equal intensities, such that intensity at those points is $\frac{7}{9}$th of maximum intensity.	(S) $\frac{3D\lambda}{d}$
	(T) $\frac{2D\lambda}{3d}$

Ans. A $\rightarrow$ Q, R, S;
B $\rightarrow$ P, Q, R, S, T;
C $\rightarrow$ Q, R, S
D $\rightarrow$ P, Q, R, S

Sol. Initially at a distance x from central maxima on screen is

$$I = I_0 + 4I_0 + 2\sqrt{I_0}\sqrt{4I_0}\cos\frac{2\pi x}{\beta}, \text{where} \beta = \frac{D\lambda}{d}$$

$I_{max} = 9I_0$ and $I_{min} = I_0$

(A) At points where intensity is $\frac{1}{9}$th of maximum intensity, minima is formed

$\therefore$ Distance between such points is $\beta, 2\beta, 3\beta, 4\beta, ...$

(B) At points where intensity is $\frac{3}{9}$th of maximum intensity, $\cos\frac{2\pi x}{\beta} = -\frac{1}{2}$ or $x = \frac{\beta}{3}$

$\therefore$ Distance between such points is $\frac{\beta}{3}, \frac{2\beta}{3}, \beta, \beta+\frac{\beta}{3}, \beta+\frac{2\beta}{3}, 2\beta, ...$

(C) $\cos\frac{2\pi x}{\beta} = 0$ or $x = \frac{\beta}{4}$

$\therefore$ Distance between such points is $\frac{\beta}{2}, \beta, \beta+\frac{\beta}{2}, 2\beta, ...$

(D) $\cos\frac{2\pi x}{\beta} = \frac{1}{2}$ or $x = \frac{\beta}{6}$

$\therefore$ Distance between such points is $\frac{\beta}{3}, \frac{2\beta}{3}, \beta, \beta+\frac{\beta}{3}, \beta+\frac{2\beta}{3}, 2\beta, ...$

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

11. A box containing an inductor of inductance L, a capacitor of capacitance C and a resistor of resistance R. When 250 V d.c. is applied to the terminals of the box, a current of 1.0 plus an oscillating current flows in the current. When an a.c. source of 250 V (r.m.s) at 2250 rad/sec is connected, an rms current of 1.25 A is produced. This current rises with frequency and becomes maximum at 4500 rad/sec. If $2R\sqrt{CL}$ is x SI unit then 9x is ____

Ans. 1

Sol. The constant + oscillating current shows that the L and C are in series, and this combination the L and C are in series, and this combination is in parallel with R = 250 ohm.

$$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$\frac{1.25}{250} = \left|\frac{1}{250} + \frac{1}{i\omega L + \frac{1}{i\omega C}}\right|$$

$$= \left| \frac{1}{250} + \frac{i\omega C}{1+\omega^2 LC} \right|$$

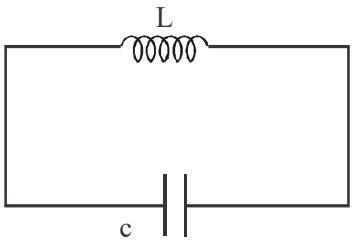
$$= \left| \frac{1}{250} + \frac{i2250C}{1-0.25} \right|$$

Hence:

$4500^2 LC = 1$

Hence $L = \frac{4}{81} H$.

12. In L-C circuit shown in the figure, the value of current at t = 0 is i_0 and charge on the capacitor at the same moment is $i_0\sqrt{3LC}$. The value of the maximum current through the inductor is $\frac{n}{2} i_0$, then n is

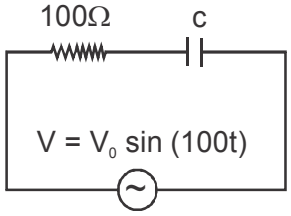


Ans. 4

Sol. $\frac{3LC}{2C} i_0^2 + \frac{1}{2} L i_0^2 = \frac{1}{2} L i^2$

$i = 2i_0$

13. A series R-C circuit is connected to an AC supply as shown in the figure. If power factor of the circuit is $\frac{1}{\sqrt{5}}$, then the value of the capacitance C $(\text{in } \mu F)$ is $\frac{1000}{n}$ then n is



Ans. 5

Sol. $\cos\phi = \frac{R}{z} = \frac{100}{\sqrt{100^2 + x_C^2}} = \frac{1}{\sqrt{5}}$

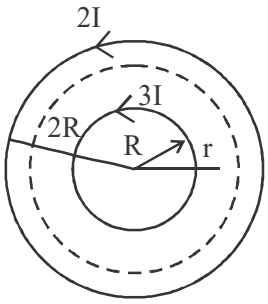
$100^2 \times 5 = 100^2 + x_c^2$

$x_C^2 = 4 \times 100^2$

$\frac{1}{c \times 100} = 2 \times 100$

$c = 200\mu F$

14. Two long solenoids are arranged coaxially, there radii are R and 2R, there coils have same number of turns per unit length. Current starts from 0 and increase at the rate 3k and 2k (in both coils in same sense) inner and outer coils respectively. As a result of increasing currents charged particle initially at rest between the solenoids it starts moving along a circular trajectory of radius [root (a/b)] then find (a + b).



Ans. 5

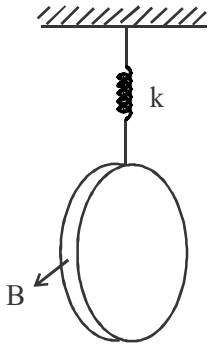
Sol. $\left[\dfrac{\mu_0 n3k\pi R^2+\mu_0 n2k\pi r^2}{2\pi r}\right]\dfrac{q}{m}t_m=\mu_0 n2Iqr$

$3R^2+2r^2=4r^2$

$2r^2=3R^2$

$r=\sqrt{\dfrac{3}{2}}R$

15. A conducting disc of mass 'm' and volume 'V' is suspended with the help of a light spring of force constant 'k' from a fixed support. Thickness of the disc is much smaller than its radius. A uniform magnetic field of induction 'B' parallel to the plane of the disc is established. The disc is pulled down by very small distance b from equilibrium position and then released. $\left(B^2V\varepsilon_0=3M\right)$. Then maximum velocity of the disc is $\dfrac{b}{n}\sqrt{\dfrac{k}{m}}$. Find n.



Ans. 2

Sol. $K\left(x_0+x\right)-mg-B^2ct_0at_0$

$kx-B^2ct_0^2a=ma$

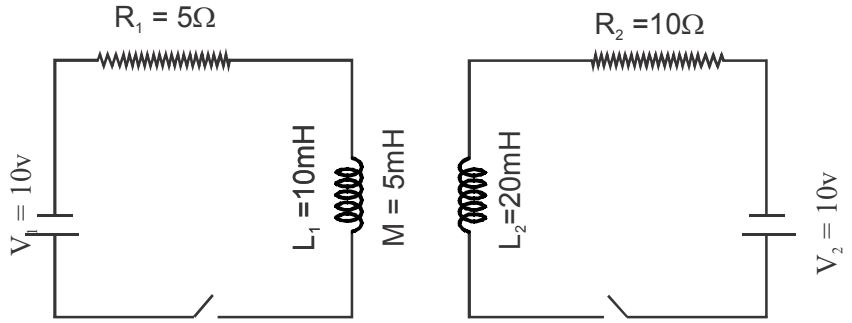
$a=\dfrac{kx}{B^2ct_0^2+m}$

$Bvt_0=\dfrac{Q}{C}$

$Q=BCVt_0$

$i=BCt_0a$

16. The inductors of two LR circuits are placed next to each other, as shown in the figure. The values of self-inductance of the inductors, resistances, mutual-inductance and applied voltages are specified in the given circuit. After both the switches are closed simultaneously, the total work done by the batteries against the induced EMF in the inductors by the time the currents reach their steady values is x mJ then $x/5$ is _____

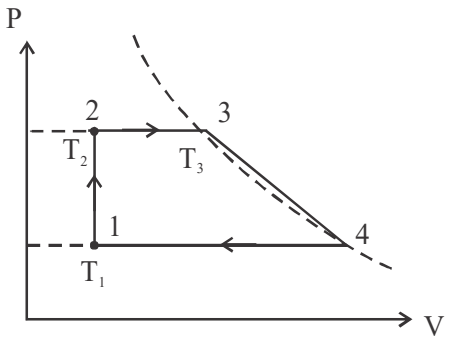


Ans. 8

Sol. $\frac{1}{2}10^{-3}\times10\times4+\frac{1}{2}\times20\times10^{-3}4-5\times10^{-3}\times4$

$(20+40-20)10^{-3}$

17. A cyclic process (cycle) 1-2-3-4-1 consisting of two isobars 2-3 and 4-1, isochor 1-2, and a certain process 3-4 represented by a straight line on the P-V diagram (see Fig.) involves n moles of an ideal gas. The gas temperatures in states 1, 2 and 3 are T_1, T_2 and T_3 respectively and points 2 and 4 lie on the same isotherm, the work done by the gas during the cycle is $nR(T_2-T_1)\left(\frac{T_2}{T_1}+\frac{T_3}{T_2}-\frac{4}{a}\right)$, then 'a' is



Ans. 2

Sol. $\frac{1}{2}(V_3-V_2+V_4-V_1)(P_2-P_1)$

express V_2, V_3, V_4 in terms of V_1

18. Following reactions are known as inverse beta decay

$p + \bar{\nu} \rightarrow n + e^+$

$n + \nu \rightarrow p + e^-$

These reactions have extremely low probabilities. Because of this, neutrinos and antineutrinos are able to pass through vast amount of matter without any interaction. In an experiment to detect neutrinos, large number of neutrinos coming out from beta decays of a radioactive material were made to pass through a tank of water, containing a cadmium compound in solution, which provided the protons to interact with antineutrinos. Immediately after a proton absorbed a neutrino to yield a positron and a neutron, the positron encountered an electron and both got annihilated. The gamma ray detectors surrounding the tanks responded to the resulting photons. This confirmed that the above reaction has taken place. The maximum energy of this photon is x Mev then 10x is (take the mass of an electron to be 0.000537 u.)

Ans. 5

PART-2 : CHEMISTRY

SECTION–I : (Maximum Marks : 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –1	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

1. Select the correct statements regarding salt solution of weak acid & strong base

(A) Solution will be neutral

(B) Degree of hydrolysis increase with increase in dilution

(C) Shows anionic hydrolysis

(D) Phenolphthalein is suitable indicator during titration of weak acid & strong base

Ans. BCD

Sol. $pH = 7 + \frac{1}{2}[pK_a + \log C]; h = \sqrt{\frac{K_w}{K_a.C}}$

2. Which of the following statements about anhydrous aluminium chloride is/are incorrect:

(A) It exists as $AlCl_3$ molecule

(B) It is a strong Lewis base

(C) It sublims at 100°C under vacuum

(D) It is not easily hydrolysed

Ans. ABD

Sol. Fact

3. For the cell

$Cu \mid CuSO_4 \parallel HCl \mid H_2 \mid Pt$

E_{cell} can be increased by -

(A) By adding NH_3 in anodic half cell

(B) Increasing partial pressure of H_2 gas

(C) By diluting $CuSO_4$ solution

(D) By diluting HCl solution

Ans. AC

Sol. $E = E^{\circ} - \frac{0.06}{2}\log\frac{\left[Cu^{2+}\right].P_{H_2}}{\left[H^{+}\right]^2}$

4. Which of the following property of lead is due to the consequence of **inert pair effect** :

(A) Higher stability of Pb^{4+} as compared to Pb^{2+}

(B) Higher stability of Pb^{2+} as compared to Sn^{2+}

(C) Size of Pb^{2+} is 133 pm but size of Pb^{4+} is 84 pm

(D) Pb^{4+} has oxidising property

Ans. BD

Sol. Facts about inert pair effect

5. Which of the following are true for an acid-base titration ?

(A) Indicators catalyse the acid base reactions by releasing or accepting H^+ ions.

(B) Indicators do not significantly affect the pH of the solution to which they are added.

(C) Acid-base reactions do not occur in absence of indicators.

(D) Indicators have different colours in dissociated and undissociated forms.

Ans. BD

Sol. Fact about indicator

6. Select the correct statement(s)

(A) Double chain silicates are known as amphiboles.

(B) In cyclic silicates two oxygen atoms per tetrahedron are shared.

(C) Orthosilicates contain discrete $\left(SiO_4\right)^{4-}$ units.

(D) Talc and mica are 3D silicates

Ans. ABC

Sol. Refer silicates from module.

7. $E^{\circ}_{A^+/A} = 0.5V$; $E^{\circ}_{B^+/B} = 0.2V$

$E^{\circ}_{C^+/C} = -0.3V$; $E^{\circ}_{D^+/D} = -0.1V$

Based on above data select correct statement under standard condition -

(A) Best oxidising agent is A^+

(B) Best reducing agent is D

(C) C can displace that D from it's solution

(D) In galvanic cell obtained by electrodes A & B current flow from B to A in external circuit

Ans. AC

Sol. Order of loosing electron : C > D > B > A

8. Buffer solution A of a weak monoprotic acid and its sodium salt in the concentration ratio x : y has pH = $(pH)_1$. Buffer solution B of the same acid and its sodium salt in the concentration ratio y : x has pH = $(pH)_2$. If $(pH)_2$— $(pH)_1$ =1 unit and $(pH)_1 + (pH)_2 = 9.5$ units, then

(A) $pK_a = 4.75$ (B) $\frac{x}{y} = 2.36$

(C) $\frac{x}{y} = 3.162$ (D) $pK_a = 5.25$

Ans. AC

Sol. $pH_1 = pK_a + \log\frac{y}{x}$

$pH_2 = pK_a + \log\frac{x}{y}$

$pH_1 + pH_2 = 9.5 = 2pK_a \quad \Rightarrow pK_a = 4.75$

$pH_2 - pH_1 = 1 = 2\log\frac{x}{y} \quad \Rightarrow \log\frac{x}{y} = \frac{1}{2}$

$\frac{x}{y} = \sqrt{10} = 3.162$

SECTION–IV : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II.**
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II.**
- One or more entries in **Column-I** may match with one or more entries in **Column-II.**
- The ORS contains a 4 × 5 matrix whose layout will be similar to the one shown below :

(A) (P) (Q) (R) (S) (T)

(B) (P) (Q) (R) (S) (T)

(C) (P) (Q) (R) (S) (T)

(D) (P) (Q) (R) (S) (T)

- For each entry in **column-I,** darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks	: +2	If only the bubble(s) corresponding to all the correct match(es) is (are) darkened
Zero Marks	: 0	In none of the bubbles is darkened
Negative Marks	: 0	In all other cases

9. Match the following

Column I	**Column II**
(A) Negative charge on the anion is equal to the number of terminal oxygen atoms	(P) $Si_6O_{18}^{-12}$
(B) Three shared corners per each central atom	(Q) SiO_4^{4-}
(C) Silicon atom(s) is/are present at the center of geometry and every oxygen atom is present at each corner of the geometry	(R) $Si_4O_{10}^{-4}$
(D) Non-planar geometry	(S) $Si_2O_7^{6-}$
	(T) None of the above mentioned silicate

Ans. A - PQRS, B - R, C - PQRS, D - PQRS

Sol. Refer structures of silicates under group-14.

10. Match the following

Column I	**Column II** (Suitable indicator at end point)
(A) Strong acid vs strong base titration	(P) Methyl orange (3.1 – 4.4)
(B) Weak acid vs strong base titration	(Q) Methyl red (4.2 – 6.3)
(C) Strong acid vs weak base titration	(R) Phenolphthalein (8.3 – 10)
(D) Weak acid vs weak base titration	(S) None of the above mentioned indicator is suitable
	(T) All of the above mentioned indicators are suitable

Ans. A-PQRT, B-R, C-PQ, D-S

Sol. A – Solution is neutral, so all indicators work.
B – Solution is basic, so phenolphthalein works
C – Solution is acidic, so methyl orange and methyl red work
D – pH may be 7/<7/<7 and no indicator works here.

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

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11. In the compound $Na_2\left[B_4O_5(OH)_4\right].8H_2O$, if the

(i) Number of B—O—B bonds is 'x'
(ii) Number of B—B bonds is 'y'
(iii) Number of sp^2 hybridized B atoms is 'z'

Calculate the value of x + y + z.

Ans. 7

Sol. $x = 5, y = 0, z = 2$. Refer to the structure of borax

12. How many litre of 0.3 M NaOH solution should be added to a solution containing 0.5 moles of HCl in hydrogen electrode, to make its Oxidation potential = 0.042 V at 25°C. $\left(\log 5 = 0.7, \frac{2.303RT}{F} = 0.06V\right)$

Ans. 1

Sol. $E = 0 - \frac{0.06}{2}\log\frac{\left[H^+\right]^2}{p_{H_2}}$

$0.042 = -0.06\log\frac{\left[H^+\right]}{1}$

$\log\left[H^+\right] = -\frac{0.042}{0.06} = -0.7 = \log\frac{1}{5}$ or $\left[H^+\right] = \frac{1}{5} = 0.2$

13. 0.1 M of HA is titrated with 0.1 M NaOH, calculate the pH at end point.

Given $K_a(HA) = 5\times10^{-6}$ and $\alpha << 1$.

Ans. 9

Sol. $pH = 7 + \frac{1}{2}\left[pK_a + \log c\right]$

$= 7 + \frac{1}{2}\left[6 - \log 5 + \log 0.05\right]$

$= 7 + \frac{1}{2}\left[6 - \log 5 - \log 2 - \log 10\right]$

$= 9$

14. Find number of reagents which produce CO_2 when reacted with carbon :

dil. H_2SO_4, conc. H_2SO_4, dil. HNO_3, conc. HNO_3, conc. NaOH, Lime water,

Ans. 2

Sol. dil. HNO_3, conc. HNO_3

15. Based on the data given at 25°C for half cell $M^{(x+n)+}, M^{x+}$ / Pt

% Reduced form (M^{+x})	50	20
Potential (V)	0.1	0.112

The value of n is -

Ans. 3

Sol. $M^{(x+n)+} + ne^- \longrightarrow M^{x+}$

Case-1 50 50

$E = E^\circ - \frac{0.06}{n}\log\frac{50}{50} \Rightarrow 0.1 = E^\circ$

Case-1 80 20

$E = E^\circ - \frac{0.06}{n}\log\frac{20}{80} \Rightarrow 0.112 = 0.1 - \frac{0.06}{n}\log\frac{1}{4}$

$0.06\log 4 = 0.012 \Rightarrow \frac{0.06}{n}\times 0.6 = 0.012$

$n = \frac{6\times6}{12} = 3$

16. The solubility of $Pb(OH)_2$ in water is 5×10^{-6} M. Calculate the pH of buffer solution in which solubility of $Pb(OH)_2$ is 5×10^{-4} mol litre^{-1}.

Ans. 8

Sol. $Pb(OH)_2 \rightleftharpoons Pb^2 + 2OH^-$

s 2s

$Ksp = 4s^3 = 4\times\left(5\times10^{-6}\right)^3$

$= 4\times125\times10^{-18} = 500\times10^{-18} = 5\times10^{-16}$

$Pb(OH)_2 \rightleftharpoons Pb^{2+} + 2OH^-$

s' 2s'+ C

$Ksp = s'(2s'+C)^2 \Rightarrow 5\times10^{-16} = s'.C^2$

$C^2 = \frac{5\times10^{-16}}{s'} = \frac{5\times10^{-16}}{5\times10^{-4}} \Rightarrow C^2 = 10^{-12} \Rightarrow C = 10^{-6}$

or $\left[OH^-\right] = 10^{-6} \quad \therefore \left[H^+\right] = 10^{-8} \Rightarrow pH = 8$

17. For the cell : $Pt|Hg(l)\,|\,Hg_2Cl_2(s)\,|\,KCl(1M)\,||\,H^+\,|Q|\,H_2Q|\,Pt$
$E_{cell} = 0.1239$ V at 298K, hence pH of cathodic compartment is.
Given : $E^\circ_{Calomel,RP} = 0.28V$, $E^\circ_{Q/H_2Q} = +0.699V$

Ans. 5

Sol. $E = E^\circ - \frac{0.059}{n}\log\frac{1}{\left[H^+\right]}$

$0.1239 = (0.699-0.28) + \frac{0.059}{1}\log\left[H^+\right]$

$\log\left[H^+\right] = -5$ or $pH = 5$

18. An acidic aqueous solution contains 0.02 M – $FeCl_2$ and 0.05 M – $FeCl_3$, at 25°C. Now NaOH solution is added is this solution, drop by drop. The pH range in which only the precipitation of $Fe(OH)_3$ occur with out precipitation of $Fe(OH)_2$ differs by 'x' unit. Find magnitude of 'x' ?
Given : K_{sp} of $Fe(OH)_2 = 8\times10^{-16}$
K_{sp} of $Fe(OH)_3 = 4\times10^{-28}$

Ans. 2.00

Sol. $\left[OH^-\right]_{min}$ to ppt $Fe(OH)_2 = \left(\frac{8\times10^{-16}}{0.02}\right)^{\frac{1}{2}} = 2\times10^{-7}$

$\therefore P^H_{min} = 7.3$

$[OH^-]_{min}$ to ppt $Fe(OH)_3 = \left(\frac{4\times10^{-28}}{0.05}\right)^{\frac{1}{3}} = 2\times10^{-9}$

$\therefore P^H_{min} = 5.3$

$\therefore$ DpH = 7.3 – 5.3 = 2.0 unit

PART-3 : MATHEMATICS

SECTION–I : (Maximum Marks : 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –1	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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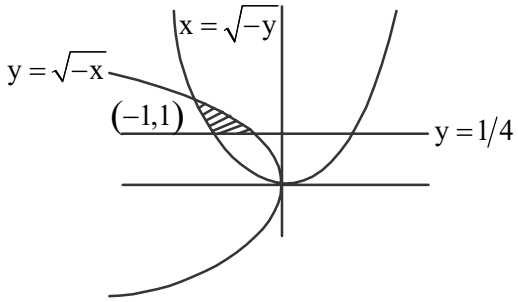
1. If area bounded by the curves $y \le \sqrt{-x}, x \ge -\sqrt{y}$ and $y \ge \frac{1}{4}$ is $\frac{1}{3}\left(\frac{m}{n}\right)^2$, where m & n are coprime numbers, then which of the following is (are) true-

(A) $|m-n|=1$ (B) m is prime & n is even

(C) n is prime number and m is even number (D) m + n is an odd number

Ans. A, B, D

Sol.



$y \le \sqrt{-x}$ & $x \ge -\sqrt{y}$ & $y \ge \frac{1}{4}$

Required Area

$$= \int_{1/4}^{1} \left(-y^2 + \sqrt{y}\right) dy = \left(-\frac{y^3}{3} + \frac{2}{3}y^{3/2}\right)_{1/4}^{1}$$

$$= \left(-\frac{1}{3} + \frac{2}{3}\right) - \left(-\frac{1}{3.64} + \frac{2}{3}.\frac{1}{8}\right)$$

$$=\frac{1}{3}+\frac{1}{3.64}-\frac{1}{3}.\frac{1}{4}=\frac{1}{3}\left(1+\frac{1}{64}-\frac{1}{4}\right)$$

$$=\frac{1}{3}\left(\frac{64+1-16}{64}\right)=\frac{1}{3}\left(\frac{7}{8}\right)^2=\frac{1}{3}\left(\frac{m}{n}\right)^2$$

$\therefore$(A) $|m-n|=1$

(B) $m=7, n=8$

(C) $m+n=7+8=15$

2. The graph of the function y = f(x) passing through the point $(-\pi,1)$ and satisfying the differential equation $\cos^2 x.\frac{dy}{dx}+y=1+\cos^2 x\, e^{-\tan x}$ then-

(A) $f\left(\frac{\pi^-}{2}\right)=1$

(B) $f\left(\frac{\pi^+}{2}\right)$ approaches to infinity

(C) $f'(0)=0$

(D) $f'(0)=1-\pi$

Ans. A, B, D

Sol. $\frac{dy}{dx}+\sec^2 x\, y=\left(\sec^2 x+e^{-\tan x}\right)$

$$\text{I.F.}=e^{\int \sec^2 x dx}=e^{\tan x}$$

$$y.e^{\tan x}=\int\left(\sec^2 x+e^{-\tan x}\right)e^{\tan x}dx$$

$$\Rightarrow y.e^{\tan x}=\int\left(\sec^2 x.e^{\tan x}+1\right)dx$$

$$\Rightarrow y.e^{\tan x}=e^{\tan x}+x+c$$

$$\Rightarrow y=1+e^{-\tan x}+ce^{-\tan x}$$

$\therefore$ at $x=-\pi, y=1$

$$\Rightarrow c=\pi$$

$$y=1+(x+\pi)e^{-\tan x}$$

$$\therefore f\left(\frac{\pi^-}{2}\right)=1,\ f\left(\frac{\pi^+}{2}\right)=\infty$$

$$\& \ f'(x)=(x+\pi)e^{-\tan x}\left(-\sec^2 x\right)+e^{-\tan x}$$

$$f'(0)=-\pi+1$$

3. If $\vec{r}$ is a non-zero vector such that $\vec{r}.\vec{a}=\vec{r}.\vec{b}=\vec{r}.\vec{c}=1$ ($\vec{a},\vec{b},\vec{c}$ are noncoplanar) then-

(A) If $A(\vec{a}),B(\vec{b}),C(\vec{c})$ are the vertices of a ΔABC, then its area is $\left|\frac{\vec{r}}{2}\right|\left[\vec{a}\ \vec{b}\ \vec{c}\right]$

(B) If $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular unit vectors then $|\vec{r}|=3$

(C) If $A(\vec{a}),B(\vec{b}),C(\vec{c})$ are vertices of ΔABC, then its area is $\left|\frac{\vec{r}}{2}\right|$

(D) If $\vec{a},\vec{b},\vec{c}$ are mutually perpendicular unit vectors then $|\vec{r}|=\sqrt{3}$

Ans. A, D

Sol. $\vec{a},\vec{b},\vec{0}$ are non coplanar then

$\vec{r}=x\vec{a}+y\vec{b}+z\vec{c}$

$\vec{r}.\vec{a}=x(a^2)$ (if $\vec{a},\vec{b},\vec{c}$ are mutually perpendicular)

$x=1$

similarly $y=z=1$

$\vec{r}=\vec{a}+\vec{b}+\vec{c}\Rightarrow|\vec{r}|=\sqrt{3}$

Let $\vec{r}=x(\vec{a}\times\vec{b})+y(\vec{b}\times\vec{c})+z(\vec{c}\times\vec{a})$

for option A $\vec{r}.\vec{a}=y\left[\vec{b}\,\vec{c}\,\vec{a}\right]$

$y=\frac{1}{\left[\vec{b}\,\vec{c}\,\vec{a}\right]}$

similarly $x=z=\frac{1}{[abc]}$

$|\vec{r}|\left[\vec{a}\,\vec{b}\,\vec{c}\right]=\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}$

$\text{area}=\frac{|\vec{r}|\left[\vec{a}\,\vec{b}\,\vec{c}\right]}{2}$

4. If plane P containing lines $L_1:\frac{x-1}{k}=\frac{y-2}{2}=\frac{z-3}{3}$ & $L_2:\frac{x-2}{2}=\frac{y-3}{k}=\frac{z-4}{3}$, then-

(A) L_1 & L_2 can be parallel

(B) L_1 & L_2 can be intersecting

(C) Equation of plane P can be $x-y+1=0$

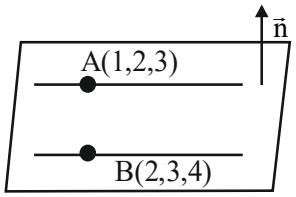
(D) Equation of plane P can be $x+y-2z+3=0$

Ans. A, B, C, D

Sol. $\begin{vmatrix}2-1 & 3-2 & 4-3\\ k & 2 & 3\\ 2 & k & 3\end{vmatrix}=0$

$\Rightarrow (k-3)^2 = 1 \Rightarrow k = 2$ or 4

for k = 2, lines are parallel

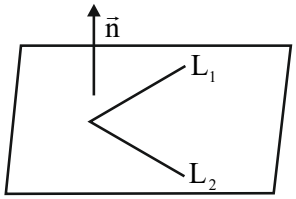


$\vec{n} = \overrightarrow{AB} \times \left(2\hat{i} + 2\hat{j} + 3\hat{k}\right) = \hat{i} - \hat{j}$

plane: $(x-1) - (y-2) + 0(z-3) = 0$

$\Rightarrow x - y + 1 = 0$

for k = 4, lines are intersecting



$\vec{n} = \vec{n}_1 \times \vec{n}_2 = -6\hat{i} - 6\hat{j} + 12\hat{k}$

plane: $1(x-1) + 1(y-2) - 2(z-3) = 0$

$x + y - 2z + 3 = 0$

5. Let n be a positive integer $f(n) = 1! + 2! + 3! + \ldots + n!$ and $P(x)$ and $Q(x)$ be polynomials in x such that $f(n+2) = P(n)f(n+1) + Q(n)f(n)$ for all $n \geq 1$, then

(A) $P(x) = x + 3$ (B) $Q(x) = -x - 2$

(C) $P(x) = -x - 2$ (D) $Q(x) = x + 3$

Ans. AB

Sol. We have $f(n+2) - f(n+1) = (n+2)!$

$= (n+2)(n+1)! = (n+2)\left[f(n+1) - f(n)\right]$

$\Rightarrow f(n+2) = (n+3)f(n+1) - (n+2)f(n)$

$\therefore P(x) = x + 3$ and $Q(x) = -x - 2$

6. The solution to the differential equation $ydx - \left(x + x^3 y\right)dy = 0$, where $y(1) = -\frac{3}{2}$ can be-

(A) $xy^2 = \frac{3}{2}$ (B) $x^2 y = \frac{3}{2}$

(C) $xy^2 = -\frac{3}{2}$ (D) $x^2 y = -\frac{3}{2}$

Ans.D

Sol. $ydx = \left(x + x^3y\right)dy$

$\Rightarrow \dfrac{dy}{dx} = \dfrac{y}{x + x^3y}$

$\Rightarrow \dfrac{dx}{dy} = \dfrac{x + x^3y}{y}$

$\Rightarrow \dfrac{dx}{dy} - \dfrac{x}{y} = x^3$

$\dfrac{1}{x^3}\dfrac{dx}{dy} - \dfrac{1}{x^2y} = 1$

put $-\dfrac{1}{x^2} = t$

$\Rightarrow \dfrac{2}{x^3}\dfrac{dx}{dy} = \dfrac{dt}{dy}$

$\Rightarrow \dfrac{dt}{dy} + 2\dfrac{t}{y} = 2$

$\text{I.F.} = e^{\int \frac{2}{y}dy} = e^{\log_e y^2} = y^2$

$\therefore t.y^2 = \int 2.y^2 dy$

$\dfrac{y^2}{-x^2} = \dfrac{2}{3}y^3 + c$

at $x = 1,\ y = -3/2 \Rightarrow c = 0$

$-\dfrac{1}{x^2} = \dfrac{2}{3}y \Rightarrow x^2y = -\dfrac{3}{2}$

7. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be the three vectors having magnitude 1, 1 and 2 respectively. If $\vec{a} \times \left(\vec{a} \times \vec{c}\right) + \vec{b} = 0$, then (Given that angle between $\bar{a}$ & $\bar{c}$ is acute)

(A) $\vec{c}.\vec{b} = 1$

(B) Angle between $\vec{a}$ and $\vec{c}$ is $\dfrac{\pi}{6}$

(C) $\vec{a}.\vec{b} = 0$

(D) $\vec{a} \times \left(\vec{b} \times \vec{c}\right) = \sqrt{3}\vec{b}$

Ans. A, B, C, D

Sol. $\vec{a} \times \left(\vec{a} \times \vec{c}\right) + \vec{b} = 0$

$\Rightarrow \left(\vec{a}.\vec{c}\right)\vec{a} - \vec{c} + \vec{b} = \vec{0}$

$\Rightarrow \left(2\cos\theta\right)\vec{a} = \vec{c} - \vec{b}$

$\Rightarrow \left(2\cos\theta\right)\vec{a}.\vec{a} = \vec{c}.\vec{a} - \vec{b}.\vec{a}$

$\Rightarrow \left(2\cos\theta\right) = 2\cos\theta - \vec{a}.\vec{b} \qquad \therefore \vec{a}.\vec{b} = 0$

$\Rightarrow \left(2\cos\theta\right)\vec{a}.\vec{b} = \vec{c}.\vec{b} - 1$

$\vec{c}.\vec{b} = 1$

$4\cos^2\theta = 4 + 1 - 2\vec{b}.\vec{c}$

$\cos^2\theta = \dfrac{3}{4} \Rightarrow \theta = \dfrac{\pi}{6}$

$\Rightarrow \vec{a} \times \left(\vec{b} \times \vec{c}\right) = \sqrt{3}\vec{b}.$

8. Solution of differential equation $\frac{xdy}{ydx}=\frac{1+x^2y^2}{1-x^2y^2}$ is-

(A) $|y|=k|x|e^{\frac{-x^2y^2}{2}}$ (B) $|y|=ke^{\frac{-x^2y^2}{2}}$

(C) $|y|=k|x|e^{\frac{x^2y^2}{2}}$ (D) $|y|=ke^{\frac{x^2y^2}{2}}$

Ans. C

Sol. $\frac{xdy-ydx}{xdy+ydx}=x^2y^2$

$\Rightarrow \frac{xdy-ydx}{xy}=xy(xdy+ydx)$

$\Rightarrow \frac{dy}{y}-\frac{dx}{x}=xyd(xy)$

$\Rightarrow \ell n\left|\frac{y}{x}\right|=\frac{x^2y^2}{2}+c$

$\Rightarrow |y|=k|x|e^{\frac{x^2y^2}{2}}$

SECTION–IV : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II.**
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II.**
- One or more entries in **Column-I** may match with one or more entries in **Column-II.**
- The ORS contains a 4 × 5 matrix whose layout will be similar to the one shown below :

(A) (P) (Q) (R) (S) (T)

(B) (P) (Q) (R) (S) (T)

(C) (P) (Q) (R) (S) (T)

(D) (P) (Q) (R) (S) (T)

- For each entry in **column-I,** darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks	:	+2	If only the bubble(s) corresponding to all the correct match(es) is (are) darkened
Zero Marks	:	0	In none of the bubbles is darkened
Negative Marks	:	0	In all other cases

9. Observe the following column:

Column I	**Column II**
(A) Area enclosed by the curve $\left(y-\sin^{-1}x\right)^2=x-x^2$ is equal to	(P) 0
(B) The area of the region represented by the expression $\sqrt{2}\le\lvert x+y\rvert+\lvert x-y\rvert\le 2\sqrt{2}$ is equal to	(Q) 1
(C) If the area bounded by $y=x^2-3$ and the line $y = ax + 2$ attains its minimum value then the parameter 'a' is equal to	(R) π
(D) If k is a positive number and the area of the region bounded by the curves $y=x-kx^2$ and $ky=x^2$ attains its maximum value then k is equal to	(S) 6
	(T) $\pi/4$

Ans. A-T, B-S, C-P, D-Q

Sol. (A) We have $-1\le x\le 1$

and $x-x^2\ge 0\Rightarrow 0\le x\le 1$, curve has two branches

$y=\sin^{-1}x\pm\sqrt{x-x^2}$ and lies

between x = 0 and x = 1

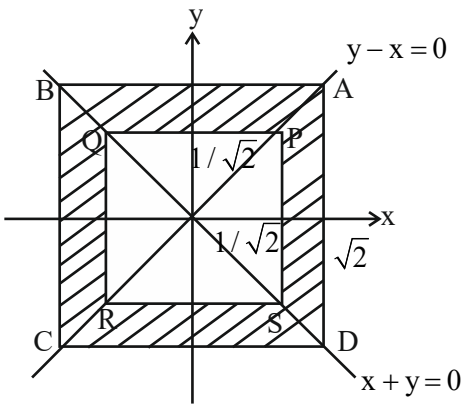
$$\therefore \text{ desired area}=\int_0^1\lvert y_1-y_2\rvert dx=2\int_0^1\sqrt{x-x^2}dx$$

$$=2\left[\frac{2x-1}{4}\sqrt{x-x^2}\right]_0^1+\frac{2}{8}\sin^{-1}(2x-1)\Big|_0^1=\frac{\pi}{4}$$

(B) If $x+y\ge 0$ and $x-y\ge 0$, then $\frac{1}{\sqrt{2}}\le x\le\sqrt{2}$

If $x+y\ge 0$ and $x-y<0$, then $\frac{1}{\sqrt{2}}\le y\le\sqrt{2}$

If $x+y<0$ and $x-y\ge 0$ then $-\sqrt{2}\le y\le-\frac{1}{\sqrt{2}}$



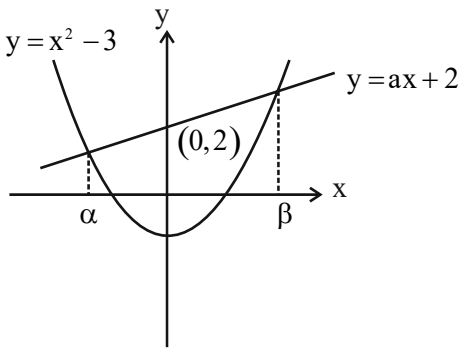
If $x+y<0$ and $x-y<0$ then $-\sqrt{2}\le x\le-\frac{1}{\sqrt{2}}$

Desired area $=\left(2\sqrt{2}\right)^2-\left(\sqrt{2}\right)^2=6$

(C) Area,

$$A=\int_{\alpha}^{\beta}\left[(ax+2)-(x^2-3)\right]dx=\left[\frac{ax^2}{2}+5x-\frac{x^3}{3}\right]_{\alpha}^{\beta}$$

$$=\beta-\alpha\left[\frac{a}{2}(\beta+\alpha)-\frac{1}{3}(\beta^2+\alpha^2+\beta\alpha)+5\right]$$



α,β are roots of $x^2-ax-5=0\Rightarrow\alpha+\beta=a,\alpha\beta=-5$

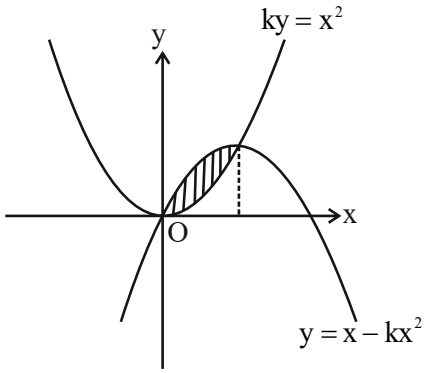
$\therefore A=\sqrt{a^2+20}\left[\frac{a^2}{6}+\frac{10}{3}\right]$, which is minimum if a = 0

(D) On solving two equations we get

$$x=0,x=\frac{k}{1+k^2}=\alpha(\text{say})$$

So, the area $A=\int_0^{\alpha}\left(x-kx^2-\frac{x^2}{k}\right)dx$

$$\Rightarrow A=\frac{\alpha^2}{2}-\frac{k\alpha^2}{3}-\frac{\alpha^3}{3k}\Rightarrow\frac{\alpha^2}{6}\left(3-2k\alpha-\frac{2\alpha}{k}\right)$$



$$=\frac{\alpha^2}{6}\left[3-\frac{2k^2}{1+k^2}-\frac{2}{1+k^2}\right]=\frac{k^2}{6(1+k^2)^2}$$

Clearly it attains its maximum when k = 1.

10. Observe the following columns:

Column I	**Column II**
(A) If $\vec{a}+\vec{b}+\vec{c}=\alpha\vec{d}$, $\vec{b}+\vec{c}+\vec{d}=\beta\vec{a}$ and $\vec{a},\vec{b},\vec{c}$ are non-coplanar then the $\left\|\vec{a}+\vec{b}+\vec{c}+\vec{d}\right\|$ is	(P) $\frac{2\pi}{3}$
(B) If $\vec{a}$ and $\vec{b}$ are unit vectors inclined at an angle θ to each other and $\left\|\vec{a}+\vec{b}\right\|<1$, then θ can be equal to	(Q) $\frac{3\pi}{4}$
(C) If $\vec{a}$ is unit vector perpendicular to another unit vector $\vec{b}$ then $\left\|\vec{a}\times\left[\vec{a}\times\left\{\vec{a}\times\left(\vec{a}\times\vec{b}\right)\right\}\right]\right\|$ is equal to	(R) $\frac{5\pi}{6}$
(D) Let $\vec{a}$, $\vec{b}$, $\vec{c}$ be three unit vectors such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$, then the angle between $\vec{a}$ and $\vec{b}$ is equal to	(S) 0
	(T) 1

Ans. A-S; B-Q,R; C-T; D-P

Sol. (A) $\vec{a}+\vec{b}+\vec{c}+\vec{d}=(\alpha+1)\vec{d}=(\beta+1)\vec{a}$

If $\alpha\neq-1$, then $\vec{d}=\left(\frac{\beta+1}{\alpha+1}\right)\vec{a}$

$\Rightarrow\vec{a}+\vec{b}+\vec{c}=\alpha\vec{d}=\alpha\left(\frac{\beta+1}{\alpha+1}\right)\vec{a}$

$\Rightarrow\left\{1-\alpha\left(\frac{\beta+1}{\alpha+1}\right)\right\}\vec{a}+\vec{b}+\vec{c}=0\Rightarrow\vec{a},\vec{b},\vec{c}$

are coplanar, which is against the given condition,

so $\alpha=-1$ and hence $\vec{a}+\vec{b}+\vec{c}+\vec{d}=\vec{0}$

(B) $\left|\vec{a}+\vec{b}\right|<1\Rightarrow\left|\vec{a}\right|^2+\left|\vec{b}\right|^2+2\left|\vec{a}\right|\left|\vec{b}\right|\cos\theta<1$

$\Rightarrow\cos\theta<-\frac{1}{2}$

So, $\frac{2\pi}{3}<\theta<\pi$

(C) $\vec{a}\times\left(\vec{a}\times\vec{b}\right)=\left(\vec{a}.\vec{b}\right)\vec{a}-\left(\vec{a}.\vec{a}\right)\vec{b}=-\vec{b}$

$a\times\left\{\vec{a}\times\left(\vec{a}\times\vec{b}\right)\right\}=\vec{a}\times-\vec{b}=\vec{a}\times\vec{b}$

$a\times\left[\vec{a}\times\left\{\vec{a}\times\left(\vec{a}\times\vec{b}\right)\right\}=\vec{a}\times\left(-\vec{a}\times\vec{b}\right)\right.$

$=\left(\vec{a}.\vec{a}\right)\vec{b}-\left(\vec{a}.\vec{b}\right)\vec{a}=\vec{b}$

(D) $\vec{a}+\vec{b}=-\vec{c}\Rightarrow\left|\vec{a}+\vec{b}\right|=\left|\vec{c}\right|^2=1$

$\Rightarrow \vec{a}.\vec{b}=-\frac{1}{2}\Rightarrow\theta=\frac{2\pi}{3}$

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

11. Each of the 'n' urns contains 4 white and 6 black balls. The (n + 1)th *urn* contains 5 white and 5 black balls. One of the (n + 1) *urns* is chosen at random and two balls are drawn from it without replacement. Both the balls turn out to be black. If the probability that the (n + 1)th *urn* was chosen to draw the balls is $\frac{1}{16}$, then the value of $\frac{n}{2}$ is equal to

Ans. 5

Sol. Let E_1 denotes the event that one of the first *n* urns is chosen and E_2 denote the event that (n + 1)th urn is selected. A denotes the event that two balls drawn are black. Then

$$p(E_1)=\frac{n}{n+1},P(E_2)=\frac{1}{n+1}$$

$$p(A/E_1)=\frac{{}^6C_2}{{}^{10}C_2}=\frac{1}{3}\text{ and }P(A/E_2)=\frac{{}^5C_2}{{}^{10}C_2}=\frac{2}{9}$$

Using Baye's Theorem, required probability

$$=P(E_2/A)=\frac{P(E_2)P(A/E_2)}{p(E_1)P(A/E_1)+P(E_2)P(A/E_2)}$$

$$\Rightarrow\frac{1}{16}=\frac{\left(\frac{1}{n+1}\right)\frac{2}{9}}{\left(\frac{n}{n+1}\right)\left(\frac{1}{3}\right)+\left(\frac{1}{n+1}\right)\left(\frac{2}{9}\right)}=\frac{2}{3n+2}\Rightarrow n=10$$

12. Let a fair coin be tossed 6 times. A, B and C three events are defined as
A : exactly 4 heads are obtained.
B : 4th head obtained on 5th toss.
C : tail is obtained, on 2nd or 4th toss

If conditional probability, $P\left(\frac{B}{A\cap C}\right)=\frac{m}{n}$ where $m,n\in N$ then find the least value of (n – m).

Ans. 7

Sol. $P\left(\frac{B}{A\cap C}\right)=\frac{P(A\cap B\cap C)}{P(A\cap C)}$

$A\cap C:(_\,T\,____)$ or $(___\,T\,__)-(_\,T\,_\,T\,__)$

$$P(A\cap C)={}^5C_4\cdot\left(\frac{1}{2}\right)^5\cdot\frac{1}{2}+{}^5C_4\cdot\left(\frac{1}{2}\right)^5\cdot\frac{1}{2}-\left(\frac{1}{2}\right)^6$$

$$=\frac{9}{2^6}$$

$A\cap B\cap C:(\underline{H}\ \underline{T}\ \underline{H}\ \underline{H}\ \underline{H}\ \underline{T})$
or $(\underline{H}\ \underline{H}\ \underline{H}\ \underline{T}\ \underline{H}\ \underline{T})$

$$P(A\cap B\cap C)=\frac{2}{2^6}$$

13. One of the body diagonal of a parallelopiped is $4\hat{j}-8\hat{k}$. If two diagonals of two different faces are $6\hat{i}+6\hat{k}$ and $-3\hat{i}+2\hat{j}-12\hat{k}$. (given all three diagonals are co-initial). If volume of the parallelopiped be 'v', then $\left[\sqrt[3]{v}\right]$, (where [.] represents greatest integer function) is

Ans. 4

Sol. Let $\vec{a},\vec{b},\vec{c}$ are edges of parallelopiped

then $\vec{a}+\vec{b}+\vec{c}=4\hat{j}-8\hat{k}$

$\vec{a}+\vec{b}=6\hat{i}+6\hat{k}$

$\vec{b}+\vec{c}=-3\hat{i}+2\hat{j}-12\hat{k}$

Solving these equation we get

$\vec{a}=3\hat{i}+2\hat{j}+4\hat{k}$, $\vec{b}=3\hat{i}-2\hat{j}+2\hat{k}$ and $\vec{c}=-6\hat{i}+4\hat{j}-14\hat{k}$

$$v=\begin{vmatrix}3 & 2 & 4\\ 3 & -2 & 2\\ -6 & 4 & -14\end{vmatrix}=120$$

$\sqrt[3]{120}\simeq 4.9=[4.9]=4$

14. Let $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=x_1\hat{i}+x_2\hat{j}+x_3\hat{k}$, where $x_1,x_2,x_3\in\{-3,-2,-1,0,1,2\}$. Number of possible vectors $\vec{b}$ such that $\vec{a}$ and $\vec{b}$ are mutually perpendicular is λ, then $(\lambda+2)/3$ is

Ans. 9

Sol. $\vec{a}.\vec{b}=0\Rightarrow x_1+x_2+x_3=0$

We have to obtain the number of integral solution of this equation $\Rightarrow$ Coefficient of x^0 in $\left(x^{-3}+x^{-2}+x^{-1}+x^0+x+x^2\right)^3$

$$=\text{Coeff. of } x^0 \text{ in } \left(\frac{1+x+x^2+x^3+x^4+x^5}{x^3}\right)^3$$

$$=\text{Coeff. of } x^9 \text{ in } (1-x^6)^3(1-x)^{-3}={}^{11}C_9-3.{}^5C_3=25$$

15. If the planes $x-cy-bz=0$, $cx-y+az=0$ and $bx+ay-z=0$ pass through a straight line, then the value of $a^2+b^2+c^2+2abc$ is:

Ans. 1

Sol. Given planes are: $x-cy-bz=0$ .....(1)

$cx-y+az=0$ .....(2)

$bx+ay-z=0$ .....(3)

Equation of plane passing through the line of intersection of plane (1) and (2) may be taken as;

$(x-cy-bz)+\lambda(cx-y+az)=0$

or $x(1+\lambda c)-y(c+\lambda)-z(-b+a\lambda)=0$ .....(4)

If plane (3) and (4) are same; then equation (3) and (4) will be identical

$$\therefore \frac{1+c\lambda}{b}=\frac{-(c+\lambda)}{a}=\frac{-b+a\lambda}{-1}$$

or $\lambda=-\frac{(a+bc)}{(ac+b)}$ and $\lambda=-\frac{(a+bc)}{(1-a^2)}$

$$\therefore \frac{-(a+bc)}{(ac+b)}=-\frac{(ab+c)}{(1-a^2)}$$

$$\Rightarrow a-a^3+bc-a^2bc=a^2bc+ac^2+ab^2+bc$$

$$\Rightarrow 2a^2bc+ac^2+a^3-a=0$$

$$\Rightarrow a(2abc+c^2+b^2+a^2-1)=0$$

$$\Rightarrow a^2+b^2+c^2+2abc=1.$$

16. If three non-zero unequal vectors $\vec{\ell}=a\hat{i}+b\hat{j}+c\hat{k}$, $\vec{m}=b\hat{i}+c\hat{j}+a\hat{k}$ & $\vec{n}=c\hat{i}+a\hat{j}+b\hat{k}$ are such that $\vec{r}.\vec{\ell}=\vec{r}.\vec{m}=\vec{r}.\vec{n}=0$ where $a,b,c\in R_0$ & $\vec{r}$ is non zero vector, then value of $\frac{a^2}{bc}+\frac{b^2}{ca}+\frac{c^2}{ab}$

Ans. 3

Sol. If $\vec{r}.\vec{\ell}=\vec{r}.\vec{m}=\vec{r}.\vec{n}=0\Rightarrow \vec{\ell},\vec{m}\&\vec{n}$ are coplanar

$$\Rightarrow\left[\vec{\ell}\ \vec{m}\ \vec{n}\right]=0\Rightarrow\begin{vmatrix}a & b & c\\ b & c & a\\ c & a & b\end{vmatrix}=0$$

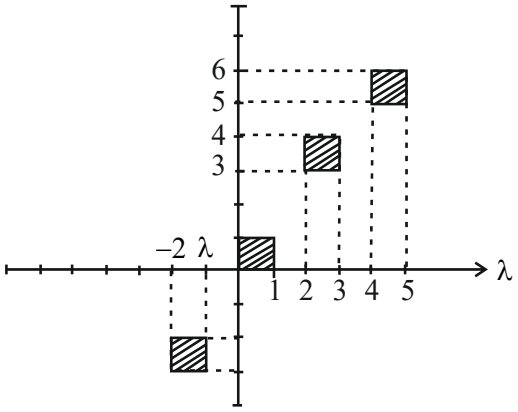
$$\Rightarrow a^3+b^3+c^3=3abc$$

$$\Rightarrow \frac{a^2}{bc}+\frac{b^2}{ca}+\frac{c^2}{ab}=\frac{a^3+b^3+c^3}{abc}=3.$$

17. P (x, y) is a point, which moves in the xy plane such that 2[y] = 3[x], where [.] denotes the greatest integer function and $-2 \le x \le 5$ and $-3 \le y \le 6$. The area of the region containing the point P(x, y) is equal to

Ans. 4

Sol. The desired region consists of four disjoint squares. So the area $= 1 \times 4 = 4$ square unit.



18. If solution y = f(x) of the differential equation, $xdy + ydx - y(x+1)dx = (x^2 + x)e^{2x}dx$ passes through (0, 0), then $\ell nf(1)$ will be

Ans. 2

Sol. $xdy + ydx - yxdx - ydx = (x^2 + x)e^{2x}dx$

$dy - ydx = (x+1)e^{2x}dx$

$\Rightarrow \frac{dy}{dx} - y = (x+1)e^{2x}$

$IF = e^{-x}$

$ye^{-x} = \int (x+1)e^{x}dx + C$

$ye^{-x} = xe^{x} + C$

$y = xe^{2x} + c.e^{x}$

c = 0 [since it passes through (0, 0)]

$y = xe^{2x}$

$\therefore f(1) = e^2 \quad \ell nf(1) = 2.$